

LABORATORIO DE ESPECIALIDADES BIOQUÍMICAS

[Specialized Biochemistry Laboratory]  
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[Translation from Spanish into English — for informational reference]

|                          |                     |                                  |
|--------------------------|---------------------|----------------------------------|
| <b>PATIENT:</b>          | <b>ADER, MANUEL</b> | <b>Accession Number: 226104</b>  |
| Patient National ID No.: | 45683086            | Date: February 6, 2026, 08:52    |
| Age:                     | 21 years            | Origin: Hospital Británico       |
| Date of birth:           | April 26, 2004      | Physician: Yelitze Choque Ayllón |
| Gender:                  | Male                |                                  |

NEXT-GENERATION SEQUENCING. MYELOID PANEL.

Sample: Bone marrow.

Clinical information / Provisional diagnosis: Acute myeloid leukemia (AML). Early relapse post allogeneic hematopoietic stem cell transplantation (allo-HSCT).

Method: Community Myeloid Solution v2 (MYS2). Target capture technology.

RESULTS:

VARIANTS WITH POTENTIAL CLINICAL SIGNIFICANCE:

| Gene (OMIM)<br>Transcript              | Variants                               | Genomic position       | Type of Variant<br>Exon | VAF<br>Depth   | dbSNP<br>COSMIC             |
|----------------------------------------|----------------------------------------|------------------------|-------------------------|----------------|-----------------------------|
| <b>WT1</b><br>(607102)<br>NM_024426.6  | c.1157C>A<br>p.(Ser386*)               | chr11:32417910<br>hg38 | Nonsense<br>Exon 7      | 32.28%<br>1721 | —<br>COSV60065297           |
| <b>FLT3</b><br>(136351)<br>NM_004119.3 | c.1790_1828dup<br>p.(Tyr597_Asn609dup) | chr13:28608228<br>hg38 | Inframe<br>Exon 14      | 13.97%<br>1171 | —<br>—                      |
| <b>BLM</b><br>(210900)<br>NM_000057.4  | c.1544dup<br>p.(Asn515Lysfs*2)         | chr15:91304139<br>hg38 | Frameshift<br>Exon 7    | 6.36%<br>1550  | rs367543043<br>COSV61921500 |

FUSION GENES:

No fusion genes were detected.

ADDITIONAL GENOMIC VARIATIONS:

Copy Number Variations (CNVs) were not detected in the evaluated genes.

INTERPRETATION:

Pathogenic sequence variants were detected in the **WT1**, **FLT3**, and **BLM** genes.

The **WT1** gene (11p13) encodes a zinc-finger-type transcription factor involved in cellular development and differentiation. Mutations in the WT1 gene are found in 7–13% of AML cases and have been associated with an adverse prognosis in some reported series. This variant had been previously detected (April 8, 2025) in a bone marrow sample obtained prior to hematopoietic stem cell transplantation.

The **FLT3** gene (13q12.2) encodes a transmembrane receptor tyrosine kinase that participates in the differentiation, proliferation, and survival of hematopoietic precursor cells. The detected variant affects the juxtamembrane domain of the protein and corresponds to a 39-bp internal tandem duplication (ITD). The presence of FLT3-ITD has been reported in 32–39% of patients with acute myeloid leukemia (AML) and in

up to 2% of cases of myelodysplastic syndrome (MDS). In AML patients, FLT3 gene mutations represent a therapeutic target.

The presence of FLT3-ITD in AML patients is associated with intermediate risk (2022 ELN AML Recommendations).

The **BLM** gene (15q26.1) encodes the BLM protein, a member of the RecQ DNA helicase family. It functions as a genomic stabilizer and is involved in the regulation of DNA replication and recombination, as well as in the homologous and non-homologous double-strand break repair pathways. Somatic BLM mutations have been reported in various types of neoplasms, with a low frequency in AML (1–2%) and a prognostic significance that is currently uncertain.

Based on current knowledge, no pathogenic variants have been detected in the remaining analyzed genes, nor has the presence of fusion genes been identified.

Correlation of these results with the clinical presentation and other complementary studies is suggested.

## REFERENCES:

1. Khoury, J.D.; Solary, E.; Abla, O.; Akkari, Y.; Alaggio, R.; Apperley, J.F.; Bejar, R.; Berti, E.; Busque, L.; Chan, J.K.C.; et al. The 5th Edition of the World Health Organization Classification of Haematolymphoid Tumours: Myeloid and Histiocytic/Dendritic Neoplasms. *Leukemia* 2022, 36, 1703–1719.
2. Döhner H, Wei AH, Appelbaum FR, Craddock C, DiNardo CD, Dombret H, Ebert BL, Fenaux P, Godley LA, Hasserjian RP, Larson RA, Levine RL, Miyazaki Y, Niederwieser D, Ossenkoppele GJ, Röllig C, Sierra J, Stein EM, Tallman MS, Tien HF, Wang J, Wierzbowska A, Löwenberg B. Diagnosis and Management of AML in Adults: 2022 ELN Recommendations from an International Expert Panel. *Blood*. 2022 Jul 7: blood.2022016867. doi: 10.1182/blood.2022016867. Epub ahead of print. PMID: 35797463.
3. Arber, Daniel A et al. "International Consensus Classification of Myeloid Neoplasms and Acute Leukemias: integrating morphologic, clinical, and genomic data." *Blood* vol. 140,11 (2022): 1200–1228. doi:10.1182/blood.2022015850.
4. Papaemmanuil E, Gerstung M, Bullinger L, Gaidzik VI, Paschka P, Roberts ND, Potter NE, Heuser M, Thol F, Bolli N, Gundem G, Van Loo P, Martincorena I, Ganly P, Mudie L, McLaren S, O'Meara S, Raine K, Jones DR, Teague JW, Butler AP, Greaves MF, Ganser A, Döhner K, Schlenk RF, Döhner H, Campbell PJ. Genomic Classification and Prognosis in Acute Myeloid Leukemia. *N Engl J Med*. 2016 Jun 9;374(23):2209–2221. doi: 10.1056/NEJMoa1516192. PMID: 27276561; PMCID: PMC4979995.

## COVERAGE AND VARIANT CALLING:

The panel can detect: SNVs, Indels, CNVs, FLT3-ITD, KMT2A-PTD, and fusion genes.

The analysis strategy used for fusion gene detection allows identification of all rearrangements involving any of the genes listed below (section B), including rare fusions or those with unknown partner genes.

The detection of copy number variations (CNVs) in the studied genes allows inference of certain numerical and structural chromosomal alterations, particularly: +8, +9, +19, -5, -7, -13, del(5q), del(7q), del(11q), del(12p), del(13q), del(17p), del(20q).

### A) Panel of analyzed genes, including the complete coding sequences (CDS) of 94 genes:

*ABL1, ANKRD26, ASXL1, ASXL2, ATM, ATRX, BCOR, BCORL1, BLM, BRAF, CALR, CBL, CBLB, CBLC, CCND2, CDKN2A, CEBPA, CHEK2, CREBBP, CSF3R, CSNK1A1, CTCF, CUX1, DDX41, DHX15, DNMT3A, ELANE, ETNK1, ETV6, EZH2, FBXW7, FLT3, GATA1, GATA2, GNAS, GNB1, HRAS, IDH1, IDH2, IKZF1, IL7R, JAK1, JAK2, JAK3, KDM6A, KIT, KMT2A, KMT2D, KRAS, LUC7L2, MPL, MSH2, MYC, MYD88, NF1, NOTCH1, NOTCH2, NPM1, NRAS, PAX5, PDGFRA, PHF6, PIGA, PML, PPM1D, PRPF8, PTEN, PTPN11, RAD21, RB1, RBBP6, RUNX1, SAMD9, SAMD9L, SBDS, SETBP1, SF3B1, SH2B3, SMC1A, SMC3, SRP72, SRSF2, STAG1, STAG2, STAT3, STAT5B, TERT, TET2, TP53, U2AF1, U2AF2, UBA1, WT1, ZRSR2.*

### B) Detection of fusions involving the following genes:

*ABL1 (1, 2, 3), BCL9 (8, 9), BCR (1, 13, 14, 15, 19), CBFB (5), DEK (2, 9), EP300 (6), ETV6 (4, 5), FGFR1 (7, 8, 9, 10), FIP1L1 (10, 11, 12, 13), JAK2 (8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19), KAT6A (15, 16), KMT2A (6, 7, 8, 9, 10, 11, 12, 21, 22, 23), MEF2D (5, 6, 7), MYH11 (28, 29, 30, 31, 32), NUP214 (6, 16, 17), NUP98 (10, 11, 12, 13), PCM1 (24, 25, 26, 28, 36), PDGFRA (11, 12) [12], PDGFRB (9, 10, 11), PICALM (16, 17, 18), PML (3, 5, 6) [6], RARA (2), RUNX1 (6), RUNX1T1 (1), SET (7), TAL1 (1, 2, 3), TCF3 (13, 14, 15, 16, 17), ZNF384 (2).*

\*(introns) [exons]

### Methodology:

1. Extraction and purification of genomic DNA from the submitted sample.
2. DNA library preparation by PCR, using the Community Myeloid v2 solution (MYS2, SOPHiA).
3. Hybridization with probes (MYS2 SOPHiA) targeting the regions of interest and capture thereof.
4. Clonal amplification and massive parallel sequencing of the selected regions on the NextSeq 2000 platform (Illumina).
5. Bioinformatic analysis of the obtained DNA sequence and of the fusion genes by comparison with the reference nucleotide sequence (GRCh38).

DNA sequence variants with a variant allele frequency (VAF) > 2%, with at least 25× coverage of the alternative allele and a minimum vertical coverage of 300×, are reported.

6. Variants are filtered using the Sophia DDM software.

7. The clinical significance of variants is determined according to the updated ACMG guidelines (PMID: 25741868), ESMO and ASCO criteria, the information available in the gnomAD, dbSNP, LOVD, and/or ClinVar databases, gene-specific databases, and the literature available at the time of the study. The interpretation of detected variants is based on the knowledge available at the date of the study.

### Study limitations:

There are certain types of alterations that cannot be identified by this test:

- Relevant variants in genes or regions not included in this panel.
- Variants below the established sensitivity threshold (10 reads).

### Note:

- The result of this study must always be interpreted in the context of clinical and family data.

Variants described in population databases and classified as SNPs (Single Nucleotide Polymorphisms) or polymorphisms (MAF — minor allele frequency —  $\geq 0.01$  and/or included in dbSNP, ExAC, or 1000 Genomes) are not included in this report, nor are benign or likely benign variants, as, based on current

knowledge, they have no clinical impact. However, such information may be available upon medical request.

**Databases consulted:**

**HGMD:** <http://www.hgmd.cf.ac.uk/ac/index.php> • **OMIM:** <http://www.omim.org/> • **ClinVar:** <http://www.ncbi.nlm.nih.gov/clinvar/> • **LOVD:** <http://www.lovd.nl/3.0/home> • **BIC:** <http://research.nhgri.nih.gov/bic> • **COSMIC:** <http://cancer.sanger.ac.uk/cosmic> • **1000 Genomes:** <http://browser.1000genomes.org> • **EVS:** <http://evs.gs.washington.edu/EVS> • **dbSNP:** <http://www.ncbi.nlm.nih.gov/snp/> • **dbVar:** <http://www.ncbi.nlm.nih.gov/dbvar/> • **ExAC:** <http://exac.broadinstitute.org> • **VarSome:** <https://varsome.com> • **IARC TP53:** <https://p53.iarc.fr/>

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